

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2367

Mcclellan Ar 138 kV
February 03, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2367. The Interconnection Customer proposes a 79.2 MW solar photovoltaic facility interconnecting at the Mcclellan Ar 138 kV substation in the Kentucky area. The request was received on December 24, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 519 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,281,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GREENVILLE AVE-VALLIANT 138.0 kV line	SILVER BAY POWER-HOTEL RD 138.0 kV	177.7	251	9.2
UNIVERSITY-WAGON MOUND 138.0 kV	PINE BLUFF WATSON CHAPEL SUBSTATION-CARRAIZO 1201 345.0 kV	152.1	272	9.0
MAINWARING-SNOHOMISH SUBSTATION 138.0 kV L	WISSOTA-BERRY ST 138.0 kV	141.9	417	7.5
SAND SPRINGS SUBSTATION-RUSSELLVILLE E 138.0 kV	BERRY ST-NORTHFIELD 345.0 kV	139.2	357	8.8
GARDEN VILLAS-DAVIE 138.0 kV	DESOTO PLEASANT RUN-UNION 138.0 kV	137.8	180	5.5
FRONTIER-DELANO SUBSTATION 138.0 kV	MCCLELLAN AR-SPATSUM LORNEX 138.0 kV	132.9	202	29.6
ARVERNE-TYRANENA 138.0 kV	KEIFER-JACKSON-BEAVER DAM E 138.0 kV L	132.8	201	6.8
DESOTO PLEASANT RUN-UNION 138.0 kV	UNION-GARDEN VILLAS 138.0 kV	127.0	218	5.7
ELDORADO SUBSTATION 345-138 kV Auto	PINE BLUFF WATSON CHAPEL-CARRAIZO 1201 345.0 kV	125.1	544	29.5
HOTEL RD-SILVER BAY POWER 138.0 kV line	SILVER BAY POWER-DAMON 138.0 kV	124.3	534	5.3
MAMMOTH-STELCO 138.0 kV	PINE BLUFF WATSON CHAPEL-CARRAIZO 1201 345.0 kV	123.5	199	7.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PINE BLUFF WATSON CHAPEL-CARRAIZO 1201 345.0 kV	LEE COUNTY SOLID WASTE ENERGY-COOPER 345.0 kV L	122.7	1,468	38.2
RANCHO CARMEL-RUSSELLVILLE E 138.0 kV	NORTHFIELD-BERRY ST 345.0 kV	122.0	414	8.8
PRAIRIE HORIZON AGRI ENERGY-THORNTWN 138.0 kV	CARRAIZO 1201-PINE BLUFF WATSON CHAPEL 345.0 kV line	121.4	388	10.3
DAVIE-OAKDALE POWER STA 138.0 kV	UNION-DESOTO PLEASANT RUN 138.0 kV	120.8	196	5.5
ANSON N SUBSTATION-BOZEMAN S 138.0 kV	LEE COUNTY SOLID WASTE ENERGY SUBSTATION-COOPER 345.0 kV line	120.6	353	8.6
SAND SPRINGS-SNOHOMISH 138.0 kV	BERRY STREET-WISSOTA 138.0 kV	120.3	500	7.5
UNION-GARDEN VILLAS 138.0 kV Line	UNION-DESOTO PLEASANT RUN SUBSTATION 138.0 kV Line	118.4	226	5.5
MURPHYS-CULVER SUBSTATION 138.0 kV	PINE BLUFF WATSON CHAPEL-CARRAIZO 1201 345.0 kV	117.3	279	8.0
SANTA RITA-WALNUT CRK PP 138.0 kV	SILVER BAY POWER-HOTEL ROAD 138.0 kV	115.1	204	5.6
THORNTOWN-DEWITT 138.0 kV	CARRAIZO 1201-PINE BLUFF WATSON CHAPEL 345.0 kV	114.3	401	10.3
MISTERSKY-GUADALUPE 138.0 kV	BERRY ST-WISSOTA 138.0 kV line	113.8	305	7.7
MURPHYS-STELCO 138.0 kV	CARRAIZO 1201-PINE BLUFF WATSON CHAPEL 345.0 kV	113.6	428	11.9
GUADALUPE-MORTAN SUBSTATION 138.0 kV line	WISSOTA-BERRY ST 138.0 kV	113.4	324	7.7
MAINWARING-MORTAN 138.0 kV	WISSOTA-BERRY ST 138.0 kV	113.3	327	7.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GREENVILLE AVE-VALLIANT 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,686,000	4% (\$295,000)
WAGON MOUND-UNIVERSITY 138.0 kV	Rebuild line with higher-capacity conductor	\$10,065,000	4% (\$365,000)
SNOHOMISH-MAINWARING 138.0 kV	Rebuild line with higher-capacity conductor	\$5,456,000	2% (\$134,000)
SAND SPRINGS SUBSTATION-RUSSELLVILLE E 138.0 kV	Rebuild line with higher-capacity conductor	\$9,986,000	2% (\$227,000)
DAVIE-GARDEN VILLAS 138.0 kV	Rebuild line with higher-capacity conductor	\$14,557,000	3% (\$497,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FRONTIER-DELANO 138.0 kV	Rebuild line with higher-capacity conductor	\$6,764,000	34% (\$2,322,000)
TYRANENA-ARVERNE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,807,000	3% (\$50,000)
DESOTO PLEASANT RUN-UNION 138.0 kV	Rebuild line with higher-capacity conductor	\$15,495,000	4% (\$549,000)
ELDORADO 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$13,954,000	6% (\$763,000)
HOTEL RD-SILVER BAY POWER 138.0 kV line	Rebuild line with higher-capacity conductor	\$18,391,000	2% (\$323,000)
MAMMOTH-STELCO 138.0 kV	Rebuild line with higher-capacity conductor	\$38,438,000	6% (\$2,266,000)
PINE BLUFF WATSON CHAPEL-CARRAIZO 1201 345.0 kV	Rebuild line with higher-capacity conductor	\$61,631,000	4% (\$2,540,000)
RANCHO CARMEL-RUSSELLVILLE E 138.0 kV	Rebuild line with higher-capacity conductor	\$9,805,000	2% (\$223,000)
PRAIRIE HORIZON AGRI ENERGY-THORNTWN 138.0 kV	Rebuild line with higher-capacity conductor	\$8,312,000	4% (\$336,000)
DAVIE-OAKDALE POWER STA SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$2,276,000	3% (\$78,000)
ANSON N-BOZEMAN S 138.0 kV	Rebuild line with higher-capacity conductor	\$36,467,000	5% (\$1,858,000)
SAND SPRINGS-SNOHOMISH 138.0 kV	Rebuild line with higher-capacity conductor	\$27,580,000	2% (\$679,000)
UNION-GARDEN VILLAS 138.0 kV Line	Rebuild line with higher-capacity conductor	\$13,061,000	3% (\$445,000)
MURPHYS-CULVER 138.0 kV	Rebuild line with higher-capacity conductor	\$7,696,000	5% (\$401,000)
SANTA RITA SUBSTATION-WALNUT CREEK PP 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,533,000	5% (\$281,000)
THORNTOWN-DEWITT 138.0 kV	Rebuild line with higher-capacity conductor	\$4,154,000	4% (\$168,000)
GUADALUPE-MISTERSKY 138.0 kV	Rebuild line with higher-capacity conductor	\$19,488,000	2% (\$461,000)
MURPHYS-STELCO 138.0 kV	Rebuild line with higher-capacity conductor	\$7,615,000	5% (\$359,000)
GUADALUPE-MORTAN SUBSTATION 138.0 kV line	Rebuild line with higher-capacity conductor	\$13,290,000	2% (\$314,000)
MAINWARING-MORTAN 138.0 kV	Rebuild line with higher-capacity conductor	\$14,661,000	2% (\$347,000)
	Total allocated to project		\$16,281,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.